

Understanding LoRA Rank Trade-offs in Diffusion Model Fine-Tuning

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Abstract

Selecting the LoRA rank for diffusion model fine-tuning is a practical systems problem: higher ranks increase trainable capacity and adaptation cost, while lower ranks improve efficiency but may underfit. We present an empirical engineering study of LoRA rank trade-offs under controlled training budgets in diffusion fine-tuning. Our primary evaluation uses a DDPM U-Net backbone on CIFAR-10 with LoRA ranks $\{2, 4, 8, 16, 32\}$ under fixed optimization settings, and reports parameter counts, runtime, GPU memory, and image quality measured by FID. We then strengthen the analysis with two targeted validations: (i) an extended-budget DDPM setting (ranks 4, 8, 16; 20 epochs), and (ii) a lightweight Tiny DiT validation (ranks 4, 8, 16; 10 epochs). All experiments use a local-folder `pytorch-fid` protocol with a reproducible CIFAR-10 test reference and 2000 generated evaluation samples per rank.

Across controlled DDPM runs, moderate ranks provide the most favorable efficiency-quality trade-off: rank 4 attains the best FID (124.1380), rank 8 is competitive (124.2136), and higher ranks show diminishing returns despite larger trainable parameter budgets. In extended-budget DDPM validation, rank 16 improves only modestly ($\approx 0.68\%$ relative FID gain from 10 to 20 epochs), without overturning the moderate-rank trend. Tiny DiT validation exhibits a similar ordering, with moderate ranks again outperforming rank 16. These results support a practical, reproducible guideline: under constrained and fixed budgets, small-to-moderate LoRA ranks are often a strong first choice for diffusion fine-tuning.

1 Introduction

Diffusion models have emerged as a compelling foundation for image generation, yet end-to-end fine-tuning of these models remains computationally demanding in practical settings. Parameter-efficient fine-tuning (PEFT) methods alleviate this burden by restricting gradient updates to a small subset of parameters rather than the full backbone. Among available PEFT approaches, Low-Rank Adaptation (LoRA) is particularly attractive for diffusion fine-tuning: it is straightforward to implement, integrates naturally with modern tooling, and exposes a single interpretable hyperparameter—rank r —that directly governs adaptation capacity [6].

In practice, however, selecting an appropriate LoRA rank remains a non-trivial decision. Very low ranks may restrict representational flexibility and limit the model’s ability to adapt to new distributions, while unnecessarily large ranks increase trainable parameter counts, memory pressure, and runtime without commensurate quality improvements. Despite growing interest in diffusion model adaptation, existing literature largely emphasizes model quality and architectural scaling, offering limited controlled guidance on rank selection for reproducible fine-tuning pipelines [4, 10].

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This gap is especially consequential in resource-constrained settings where training budgets are fixed and experimental reproducibility is a hard requirement.

This paper addresses that gap through a controlled empirical study of LoRA rank trade-offs in diffusion fine-tuning. We conduct a primary sweep over ranks $\{2, 4, 8, 16, 32\}$ using a DDPM U-Net backbone on CIFAR-10, then assess trend stability through two focused validations: extended DDPM training at ranks $\{4, 8, 16\}$ over 20 epochs, and a second lightweight backbone (Tiny DiT) at ranks $\{4, 8, 16\}$ over 10 epochs. Quality is measured via FID using a reproducible local `pytorch-fid` protocol, alongside parameter efficiency, runtime, and memory metrics. Optimization settings are held fixed across all ranks to isolate the effect of rank under a controlled training budget, rather than to identify per-rank hyperparameter optima.

This work makes the following contributions:

- A controlled DDPM LoRA rank sweep on CIFAR-10 across ranks $\{2, 4, 8, 16, 32\}$ with unified training and evaluation protocols.
- Quantitative characterization of efficiency-quality trade-offs using trainable parameter counts, runtime, GPU memory, and FID as a relative optimization indicator under constrained budgets.
- Extended-budget DDPM validation (20 epochs; ranks $\{4, 8, 16\}$) examining whether higher ranks close the quality gap with additional training steps.
- A Tiny DiT second-backbone validation (10 epochs; ranks $\{4, 8, 16\}$) assessing whether rank-ordering trends generalize beyond the DDPM U-Net architecture.
- A reproducible PEFT diffusion experimentation pipeline with explicit artifact generation, scriptable validation, and structured reporting outputs.

Figure 1 provides an overview of the experimental workflow and the primary findings of the study. The results show that moderate LoRA ranks consistently provide the strongest efficiency-quality trade-off under fixed optimization budgets.

The remainder of this paper is organized as follows. Section 2 reviews related work on diffusion models and parameter-efficient fine-tuning. Section 3 introduces the problem formulation and technical background. Section 4 presents the methodology and controlled experimental design. Section 5 describes the implementation details and experimental setup. Section 6 reports the results and discussion across all evaluation tracks. Finally, Sections 7 and 8 discuss limitations and conclude the paper.

2 Related Work

Denoising diffusion probabilistic models (DDPMs) establish a likelihood-based generative framework in which a neural network learns to iteratively reverse a fixed Gaussian noising process applied to training data [4]. Since their introduction, diffusion models have expanded considerably in both scale and architectural diversity, with transformer-based backbones such as the Diffusion Transformer (DiT) demonstrating strong performance on class-conditional image generation [10]. The present work does not propose a new diffusion architecture; rather, it examines practical fine-tuning behavior under fixed training budgets, using established backbones as controlled experimental platforms.

Parameter-efficient fine-tuning (PEFT) methods reduce adaptation cost by confining gradient updates to a small subset of model parameters. Early approaches introduced lightweight adapter modules inserted between frozen transformer layers [5], while prompt-based tuning methods condition frozen models through learned input representations [9]. LoRA offers a complementary formulation:

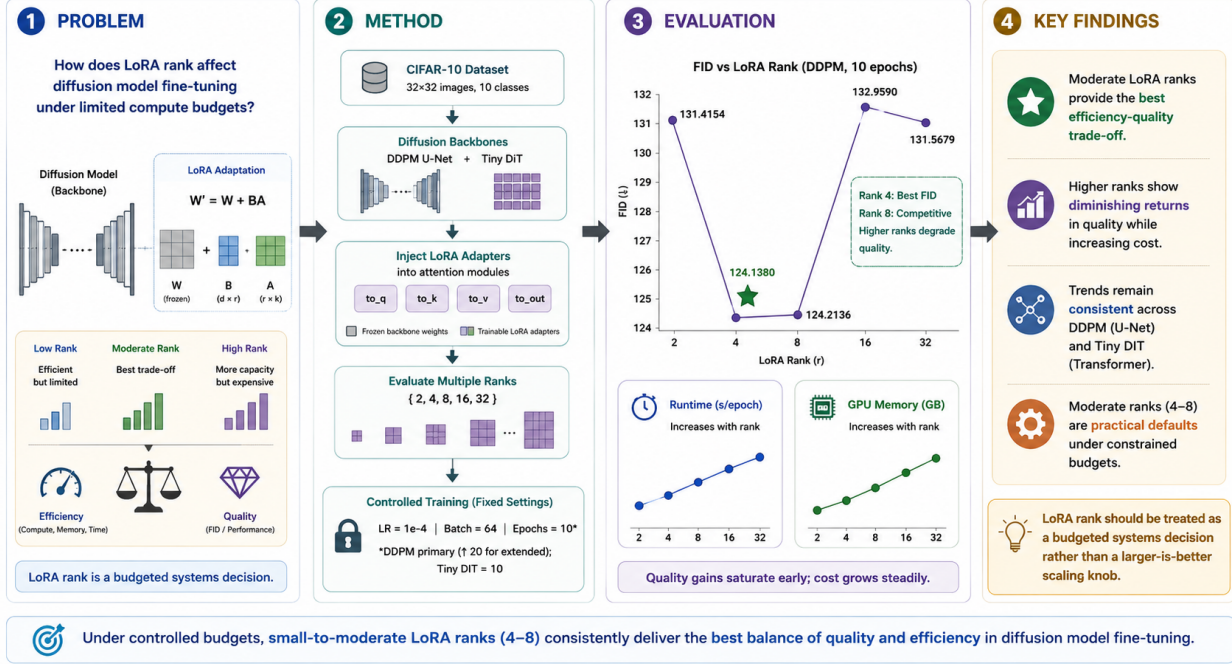


Figure 1: Overview of the controlled evaluation framework for LoRA rank trade-offs in diffusion model fine-tuning. Multiple LoRA ranks are evaluated across DDPM and Tiny DiT backbones under fixed optimization budgets, showing that moderate ranks provide the most favorable efficiency–quality balance.

rather than inserting new modules, it augments frozen weight matrices with trainable low-rank factors that can be folded back into the base weights at inference [6]. The rank parameter r provides direct, interpretable control over adaptation capacity, making LoRA a natural candidate for systematic efficiency-quality trade-off analysis under fixed compute budgets.

A growing body of work has examined the limitations of static, uniform rank assignment in LoRA. AdaLoRA [13] reparameterizes weight updates via singular value decomposition and dynamically prunes less important singular values based on layer-wise importance scores, allowing rank to adapt to the varying demands of different modules. Subsequent methods have extended this direction: ElaLoRA [1] enables both rank pruning and expansion during training via gradient-derived importance estimates, while ARD-LoRA [2] targets heterogeneous adaptation requirements across individual attention heads. Separately, rsLoRA [7] identifies a theoretical limitation in the standard LoRA scaling factor, showing that the conventional α/r normalization causes gradient magnitude to collapse as rank increases, and proposes a corrected $\alpha/\sqrt{r}$ scaling that stabilizes learning at higher ranks without additional inference cost. More recently, timestep-dependent rank adaptation has been explored for diffusion model personalization: T-LoRA [11] demonstrates that higher diffusion timesteps are disproportionately prone to overfitting, motivating rank schedules that vary with the denoising trajectory.

While these adaptive and rank-aware methods represent important advances, they introduce additional complexity—in the form of dynamic schedules, importance scoring, or modified optimization regimes—that can obscure the baseline relationship between fixed rank and adaptation quality. Our study deliberately occupies a complementary position: rather than proposing a new rank-selection mechanism, we provide controlled empirical evidence of how static rank choices

interact with training budget under uniform optimization settings. This evidence is directly relevant for practitioners who require reproducible baselines before committing to more complex adaptive strategies.

Open-source frameworks have substantially lowered the barrier to reproducible diffusion experimentation. Hugging Face Diffusers [12] provides consistent implementations of models and noise schedulers across a wide range of architectures, enabling fair cross-study comparisons. Reproducible image-quality evaluation commonly relies on the Fréchet Inception Distance (FID) [3], while CIFAR-10 remains a widely used controlled benchmark that facilitates rapid and resource-efficient comparative studies [8]. The present study builds directly on these foundations, employing a local `pytorch-fid` protocol with a fixed CIFAR-10 test reference to ensure evaluation consistency across all experimental tracks.

3 Problem Statement and Background

Selecting the LoRA rank for diffusion fine-tuning is, in essence, a budgeted systems decision: the practitioner must allocate a fixed parameter and compute budget across fine-tuning runs without per-rank hyperparameter retuning, and must do so in a way that is reproducible and transferable to similar settings. Despite the prevalence of LoRA in diffusion adaptation workflows, the empirical relationship between rank and adaptation quality under such constraints is not well characterized. This section introduces the necessary technical background and clarifies the design implications that motivate the controlled evaluation in this paper.

Denoising diffusion probabilistic models learn to generate data by reversing a fixed forward process that progressively corrupts clean samples with Gaussian noise. Formally, the model trains a neural network $\epsilon_\theta(x_t, t)$ to predict the noise injected at diffusion step t , where x_t is obtained by sampling from the forward noising process applied to a clean input x_0 [4]. At inference time, new samples are generated by iteratively applying the learned denoising network from a pure Gaussian prior, guided by a fixed noise schedule. The quality of this reverse process depends critically on the capacity of ϵ_θ to model the score function of the data distribution across all noise levels, a requirement that motivates the use of expressive U-Net and transformer-based backbone architectures [4, 10].

LoRA adapts a pretrained weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ by introducing a low-rank residual update of the form

$$W' = W + BA, \tag{1}$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are the trainable adapter matrices, and $r \ll \min(d_{\text{out}}, d_{\text{in}})$ is the rank [6]. During fine-tuning, the base parameters W are frozen and only A and B are optimized, substantially reducing the number of trainable parameters relative to full fine-tuning. The total adapter parameter count scales linearly with r : increasing rank expands the expressivity of the low-rank update but proportionally increases trainable parameter counts, memory footprint, and, to a lesser extent, runtime. Crucially, the adapted weight W' takes the same form as the original matrix, so trained adapters can be merged into W at inference with no additional computational overhead.

Because rank directly governs trainable capacity, comparing LoRA configurations across different ranks is a non-trivial experimental design problem. If optimizer settings are adjusted separately for each rank, the observed quality differences conflate the effect of rank with the effect of tuning. Our study therefore holds all optimization hyperparameters fixed across ranks, interpreting observed FID differences as evidence of rank efficiency under a controlled and uniform training budget, rather

than as indicators of per-rank hyperparameter optima. This design choice is central to the empirical framing of all results reported in Section 6.

4 Methodology

4.1 Study Design

The study is organized into three controlled experiment tracks. The primary track conducts a full DDPM rank sweep over $r \in \{2, 4, 8, 16, 32\}$ for 10 epochs, establishing the baseline efficiency-quality profile across the evaluated rank range. The second track extends the training budget of the DDPM configuration to 20 epochs for a subset of ranks $r \in \{4, 8, 16\}$, testing whether higher ranks recover quality advantages with additional optimization steps. The third track applies the same 10-epoch protocol to a lightweight Tiny DiT backbone at ranks $r \in \{4, 8, 16\}$, assessing whether the rank-ordering trends observed in the primary DDPM sweep generalize to a structurally different architecture. All three tracks use CIFAR-10 at 32×32 resolution and share identical core optimization settings: batch size 64 and learning rate 10^{-4} .

4.2 LoRA Injection and Parameter Freezing

LoRA adapters are injected into the attention projection modules of each target backbone, specifically the query, key, value, and output projections (`to_q`, `to_k`, `to_v`, `to_out.0`). All base model parameters are frozen throughout training, and only the LoRA adapter matrices are updated via gradient descent. For each experimental run, we record both total and trainable parameter counts directly from run artifacts, allowing precise verification that parameter scaling across ranks matches the theoretical expectation implied by Equation 1.

4.3 Evaluation Protocol

Generation quality is assessed using the Fréchet Inception Distance (FID) [3], computed via `pytorch-fid` against a fixed local CIFAR-10 test reference set stored at `outputs/fid_reference/cifar10_test`. To maintain evaluation consistency across all tracks and ranks, FID is computed from 2000 generated images per run. Using a local reference folder rather than a precomputed dataset statistic eliminates a common source of cross-study evaluation drift. In addition to FID, we record final training loss, wall-clock runtime, and peak GPU memory consumption from training artifacts, providing a joint view of quality and systems efficiency for each rank configuration.

4.4 Controlled-Optimization Positioning

A central design decision in this study is the deliberate choice to hold all optimization settings fixed across ranks. This means that differences in FID across ranks reflect the effect of rank under a uniform training budget, rather than the outcome of rank-specific hyperparameter search. As a consequence, results should be interpreted as controlled within-budget comparisons: they characterize which rank configurations make the most efficient use of a fixed compute allocation, not which ranks would perform best given tailored optimization. This framing is carried consistently through all experimental tracks and is fundamental to the interpretation of the findings in Section 6.

5 Experimental Setup

Table 1: Main DDPM sweep (10 epochs, CIFAR-10, fixed optimization settings).

Rank	Trainable	Trainable(%)	Runtime(s)	Peak MB	Final Loss	FID
2	49152	0.0432	521	2048.21	0.0532	131.4154
4	98304	0.0864	522	2048.89	0.0594	124.1380
8	196608	0.1727	524	2050.24	0.0587	124.2136
16	393216	0.3447	521	2052.95	0.0414	132.2549
32	786432	0.6871	528	2074.49	0.0634	131.6479

5.1 Environment and Stack

Experiments were executed on OSC via SLURM with GPU-backed jobs. The implementation stack uses PyTorch, Hugging Face Diffusers [12], PEFT/LoRA [6], and Hugging Face datasets tooling. All runs are generated through scripted pipelines that produce structured artifacts at each stage, including per-rank training logs, generated image sets, and FID evaluation outputs, supporting end-to-end reproducibility.

5.2 Data and Backbones

All experiments use CIFAR-10 [8] at 32×32 resolution as the training and evaluation dataset. The primary backbone is a DDPM U-Net [4], selected for its widespread use in controlled diffusion fine-tuning studies. The validation backbone is a lightweight Tiny DiT-style `Transformer2DModel` inspired by the DiT architecture [10], included as a cross-backbone check rather than as a large-scale benchmark. The two backbones differ in architectural family—convolutional versus transformer-based—allowing a limited assessment of whether rank-ordering trends are backbone-specific or more broadly consistent.

5.3 Training Configuration and Reported Metrics

Across all tracks, the optimizer policy is held fixed: batch size 64, learning rate 1×10^{-4} , with only rank-specific LoRA adapter parameters trainable. The main DDPM sweep and Tiny DiT validation run for 10 epochs; the extended-budget DDPM validation runs for 20 epochs. For each run, we report FID (computed via `pytorch-fid` against the local CIFAR-10 test reference using 2000 generated images), trainable parameter count and percentage of total parameters, wall-clock runtime in seconds, peak GPU memory usage in megabytes, and final training loss. This joint reporting of quality and systems metrics is intended to provide a complete view of the efficiency-quality trade-off at each rank.

6 Results and Discussion

6.1 Main DDPM Rank Sweep

Table 1 summarizes the main controlled DDPM sweep across ranks $\{2, 4, 8, 16, 32\}$. Rank 4 achieves the most favorable FID (124.1380), while rank 8 is nearly identical (124.2136). Higher ranks (16 and 32) increase trainable parameter budgets substantially but do not improve FID under the same training budget, indicating diminishing returns in adaptation quality relative to parameter cost.

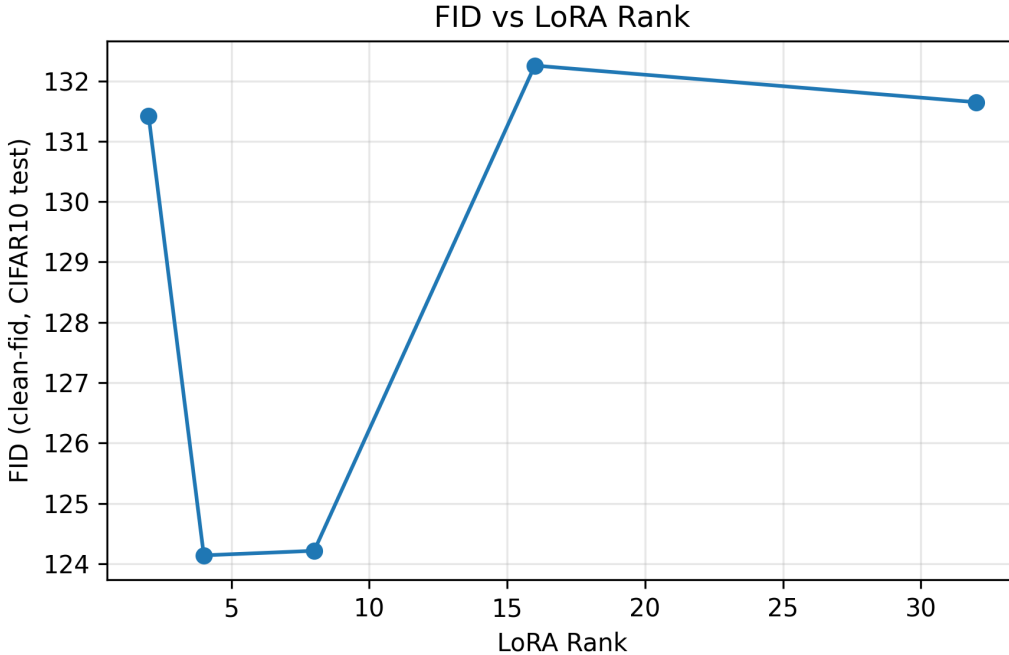


Figure 2: Main DDPM sweep: FID versus LoRA rank.

6.2 Efficiency–Quality Trade-offs

Figure 2 shows FID as a function of rank in the main sweep. The relationship is non-monotonic: quality improves from rank 2 to rank 4, plateaus at rank 8, and then degrades at ranks 16 and 32. This pattern suggests that, under the fixed optimization budget used here, moderate ranks provide the best balance between adaptation expressivity and optimization tractability. Figure 3 reinforces this point, illustrating that increasing trainable parameters does not yield monotonic FID improvement under fixed budgets.

Runtime and memory overhead scale modestly but consistently with rank (Table 1), while quality gains saturate quickly after rank 4. This asymmetry between resource cost and quality return supports an engineering recommendation to prioritize moderate ranks when training budgets are fixed and tuning time is limited.

6.3 Extended-Budget Validation

To test whether higher ranks recover quality advantages with more optimization steps, we run a 20-epoch DDPM validation on ranks $\{4, 8, 16\}$. Results are shown in Table 2. Compared with rank 16 at 10 epochs (FID 132.2549), rank 16 at 20 epochs improves to 131.3555, an absolute gain of 0.8994 and a relative improvement of approximately 0.6801%. The gain is measurable but modest, and does not reverse the broader moderate-rank preference observed under controlled budgets.

Figure 4 contrasts DDPM performance at 10 and 20 epochs for ranks 4, 8, and 16, providing a visual comparison of the modest budget-induced gains. Notably, rank 4 continues to achieve the lowest FID at 20 epochs (130.8580), further supporting the observation that moderate ranks make more efficient use of available optimization steps.

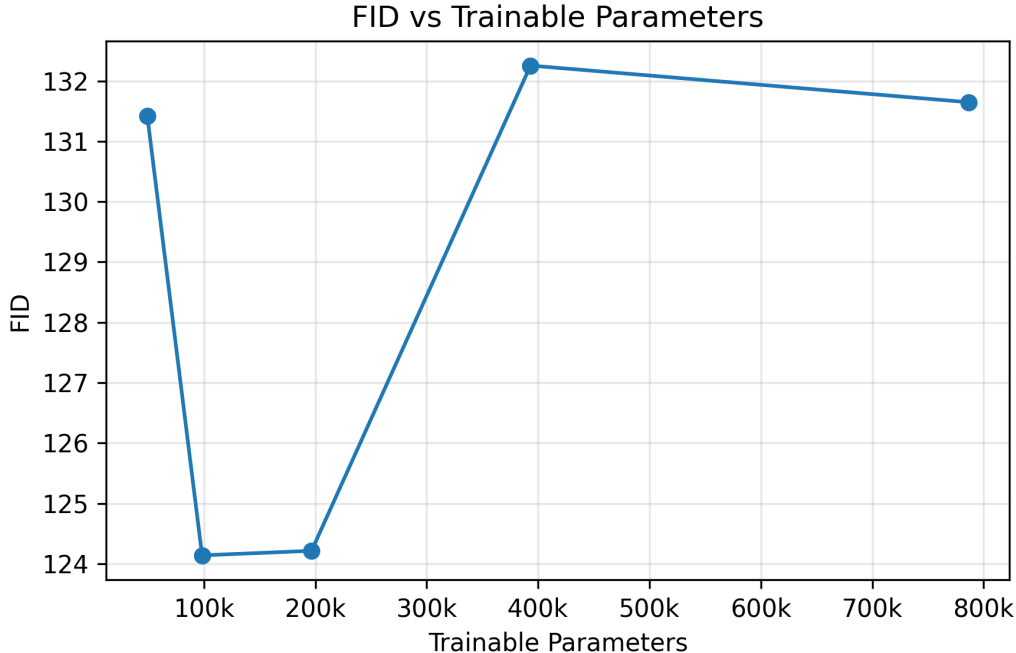


Figure 3: Main DDPM sweep: FID versus trainable parameters.

Table 2: Extended-budget DDPM validation (20 epochs, ranks 4/8/16).

Rank	Trainable	Trainable(%)	Runtime(s)	Peak MB	Final Loss	FID
4	98304	0.0864	1045	2065.01	0.0407	130.8580
8	196608	0.1727	1044	2066.37	0.0343	132.2700
16	393216	0.3447	1040	2069.07	0.0311	131.3555

6.4 Tiny DiT Validation

To assess whether the rank-ordering trends observed with the DDPM U-Net generalize to a structurally different backbone, we evaluate a lightweight Tiny DiT-style `Transformer2DModel` under the same 10-epoch protocol. Results are reported in Table 3. Rank 4 again achieves the best FID (383.2438), rank 8 remains competitive (387.8070), and rank 16 is clearly worse (402.0335) under the same budget. The rank-ordering is consistent with the DDPM findings, despite the architectural differences between the two backbones.

Figure 5 compares DDPM (20 epochs) and Tiny DiT (10 epochs) FID profiles for ranks 4, 8, and 16. Absolute FID levels are not directly comparable across backbones, as the two architectures operate at different quality baselines under the evaluated budgets. The key result is the consistency of rank ordering rather than the absolute values, which reinforces the generality of the moderate-rank preference within the controlled settings of this study.

6.5 Interpretation and Practical Implications

Taken together, the results across all three tracks indicate that LoRA rank should be treated as a budgeted systems decision rather than a “larger is always better” scaling knob. In this controlled setting, moderate ranks—especially rank 4, with rank 8 close behind—consistently provide stronger

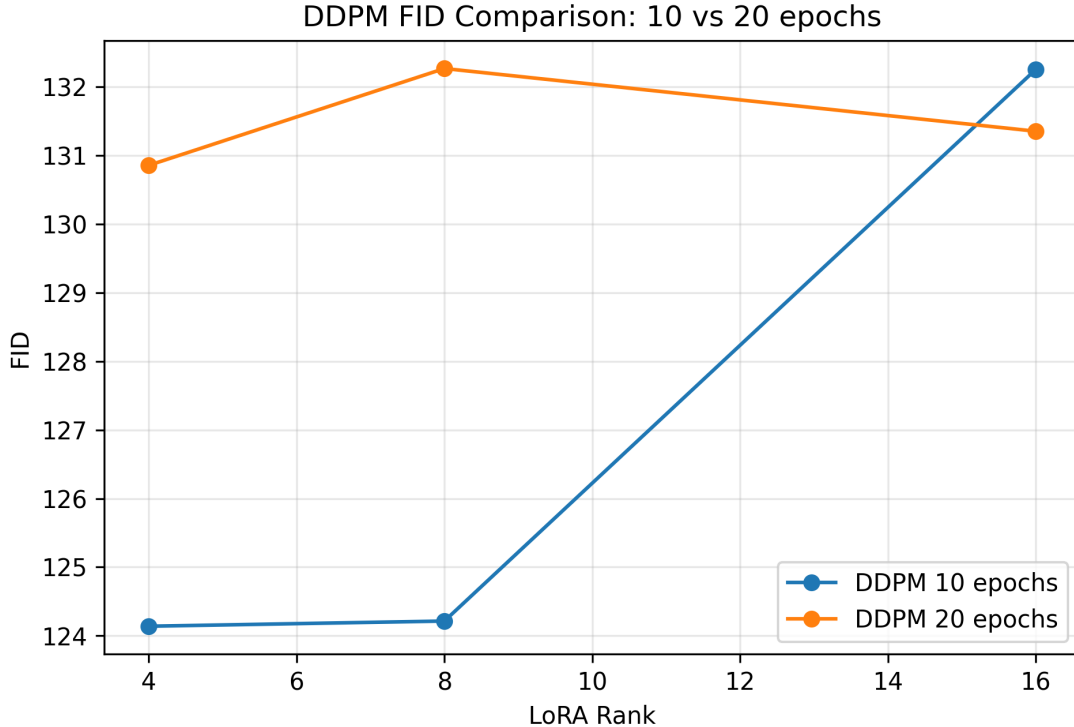


Figure 4: DDPM comparison at 10 vs. 20 epochs for ranks 4, 8, and 16.

Table 3: Tiny DiT validation (10 epochs, ranks 4/8/16).

Rank	Trainable	Trainable(%)	Runtime(s)	Peak MB	Final Loss	FID
4	16384	1.5269	549	1753.54	0.1435	383.2438
8	32768	3.0078	545	1761.75	0.1425	387.8070
16	65536	5.8400	545	1778.18	0.1347	402.0335

quality-efficiency outcomes than larger ranks, and this pattern holds across both a 10-epoch and a 20-epoch DDPM setting and across two structurally distinct backbones.

An important caveat governs the interpretation of these FID values. All models in this study were trained under deliberately constrained budgets on a low-resolution benchmark, and the generated samples remained visually undertrained under these conditions. Reported FID values should therefore be interpreted primarily as relative optimization indicators rather than as measures of final generative quality. The relevant evidence lies in the consistency of rank ordering across experimental tracks, not in the absolute FID levels achieved.

Under constrained and fixed optimization budgets, increasing rank can expand trainable capacity faster than the optimization process can utilize it, yielding weak or even negative marginal quality gains. This does not imply that high ranks are universally ineffective; rather, it suggests that larger ranks may require different optimization budgets and tuning regimes to realize their potential. Within the budget regime studied here, however, moderate ranks make more efficient use of available compute.

For practitioners tuning diffusion models under finite compute, these findings suggest starting with moderate LoRA ranks before allocating resources to large-rank sweeps, and tracking both

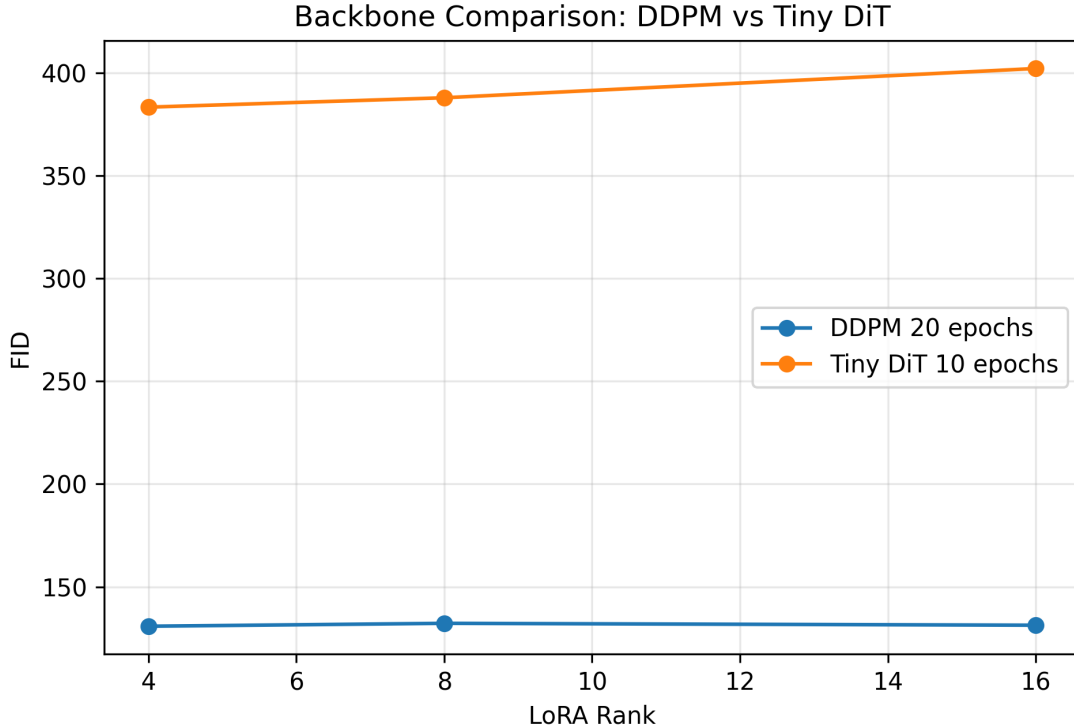


Figure 5: Backbone comparison: DDPM (20 epochs) vs. Tiny DiT (10 epochs) for ranks 4, 8, and 16.

quality and systems metrics—runtime and memory alongside FID—rather than optimizing for FID alone. The study also highlights the value of explicit artifact pipelines: rank-specific configurations, local reference data for FID, and scriptable validation steps lower the risk of hidden evaluation drift and simplify reviewer-facing reproducibility.

7 Limitations

This study is intentionally scoped as a controlled empirical engineering analysis, and several important limitations should be considered when transferring its conclusions to broader settings. All experiments use CIFAR-10 exclusively, at 32×32 resolution, which limits the generalizability of the findings to higher-resolution datasets and more diverse data distributions. The evaluated backbones—a DDPM U-Net and a lightweight Tiny DiT—are considerably smaller than the large-scale foundation models used in production diffusion systems, and rank-ordering trends may differ at scale. Training budgets and evaluation sample counts are bounded by reproducibility and resource constraints, and we do not perform rank-specific optimizer retuning or exhaustive hyperparameter search. As noted in Section 6.5, the constrained training budgets result in visually undertrained samples, and FID comparisons should be read as relative optimization evidence rather than definitive quality rankings. Finally, the results should not be interpreted as general scaling laws; they are specific to the optimization regime, architectures, and dataset examined here.

8 Conclusion

We presented a reproducible empirical study of LoRA rank trade-offs in diffusion fine-tuning under controlled training budgets. In the main DDPM sweep, moderate ranks (4 and 8) provided the most favorable quality-efficiency outcomes, while larger ranks showed diminishing returns relative to their increased trainable parameter cost. Extended-budget DDPM validation improved rank 16 only modestly, and Tiny DiT validation exhibited a consistent rank-ordering trend across a structurally different backbone.

These results support a practical recommendation for budget-constrained diffusion adaptation: small-to-moderate LoRA ranks are strong default choices when optimization settings are fixed and reproducibility is prioritized. Future work should evaluate adaptive rank allocation strategies, rank-specific optimizer tuning, larger-scale backbones, and broader datasets to determine how these trends evolve under less constrained experimental conditions.

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